(Please write your Enrollment Number) **Enrollment No._____**

MID-TERM EXAMINATION
(Course Name: BBA) (Semester: II)
(March 2025) OFF LINE mode

Subject Code: AMC 102	**Subject: Business Statistics**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	**Attempt all parts:**	**(2.5*4=10)**	**CO Mapping**
	a) Define Mode with an appropriate example.		1
	b) Define Skewness and Kurtosis briefly.		5
	c) Calculate Mean Deviation from Median for the following data on monthly income (in Rs.) of households: Income 8520 6350 7920 8360 7500		3
	d) If r= 0.8 between two variables X & Y. What does it imply? Give an example from everyday life for two such variables.		2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Calculate median from the following data: Class: Frequency 10-20: 15 20-30: 8 30-40: 17 40-50: 29 50-60: 7 60-70: 4		5
	b) Compute arithmetic mean of the following: Class: Frequency 0-10: 12 10-20: 18 20-30: 27 30-40: 20 40-50: 17 50-60: 6		4
	c) The tax authority collected the following amount of tax from different firms in a particular market. Amount of Taxes (in 000 Rs.): No. of firms 10: 3 11: 12 12: 18 13: 12 14: 3		2

	Calculate the quartile Deviation and coefficient of Quartile Deviation.		
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) A study of 1000 companies gives the following information:		3

Profit (Rs.)	No. of companies
0-10	10
10-20	20
20-30	30
30-40	50
40-50	40
50-60	30

Calculate the standard deviation of the profit earned by assumed mean method.

b) Seven students of a class secured the following marks in economics and history. Calculate the coefficient of correlation with the help of given data: — 4

Economics	History
66	58
90	76
89	65
55	58
58	53
44	49
42	56

c) Calculate the linear regression equation of advertisement cost on sales from the data given below: — 1

Advertisement Costs (in Crores)	Sales (in Crores)
2	3
4	7
6	5
8	10

Also, depict the value of sales, if the advertisement costs go upto Rs. 10 crores